Likhith Sadahalli Thammegowda

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OPENSOURCE CONTRIBUTIONS

CONTRIBUTOR TO ECLIPSE KUKSA DATABROKER (RUST)

Contributed to the Eclipse kuksa-databroker by integrating OpenTelemetry for distributed tracing. This improved system observability, enabled trace analysis across microservices, and helped identify latency bottlenecks in automotive data communication.

EXPERIENCE

ROBERT BOSCH MOBILITY — OCT 2025 – MAR 2026

***MASTER THESIS:** FAULT INJECTION MECHANISMS IN VIRTUALIZED ENVIRONMENTS AND ITS OBSERVABILITY* | *Go, Docker, Kubernetes, eBPF, Prometheus, Thanos, AWS (EC2, S3 bucket, Lambda functions), ReactJS*

- Engineered a fault injection framework for cloud nodes to analyze how resource consumption affects the vECU simulations running within them.
- Deployed cross-platform **observability exporters** (Node, Process, Windows, custom eBPF) as Kubernetes DaemonSets, utilizing kubernetes_sd_configs for dynamic discovery and label-based controls to manage resource overhead.
- Developed a **full-stack observability suite** featuring interactive fault injection and log aggregation, backed by custom APIs that feeds metrics and logs into an LLM context window to generate natural language performance insights.

ROBERT BOSCH CORPORATE RESEARCH — FEB 2025 – JULY 2025

INTERN | *RUST, Go, Docker, OpenTelemetry, Grafana, WebRTC, Microsoft Azure*

Low-latency communication strategies to enable Vehicle-to-Cloud (V2C) teleoperation. The goal is to reduce end-to-end latency in teleoperation of vehicles by evaluating and integrating WebRTC-based communication for real-time streaming and signaling.

- Evaluating **Microsoft Azure** infrastructure components including:
  - **Proximity placement groups** to ensure physical co-location of compute resources
  - Evaluating the effect of **VNet** Peering for Low-Latency Interconnects
  - Architectural Considerations for Cloud **Scalability** and **Fault Tolerance**
- Analysis of Peer-to-Peer Network Paths Across MNOs, ISPs, and Cloud Infrastructure in Wired and Wireless scenarios.
- Investigating the use of **WebRTC** to reduce round-trip latency in teleoperation use cases involving camera streams, control signals, and feedback loops.
  - Implementing SFUs in cloud to reduce peer to peer communication latency.

ROBERT BOSCH CORPORATE RESEARCH — FEB 2024 – NOV 2024

WORK STUDENT | *RUST, Go, Docker, OpenTelemetry, Jaeger, Grafana, Prometheus, eBPF, ROS2, gRPC*

Black box testing of kuksa-databroker using ghz tool. Modified the ghz tool written in **Go language** to capture custom metrics helpful in black box testing to obtain end-to-end latency of **kuksa-databroker**.
Leveraged **Prometheus** for real-time performance monitoring and used **Grafana** dashboards for visualizing the collected metrics.
Black box testing of kuksa-databroker was used to verify the advantage of L4S (Low Latency, Low Loss, Scalable Throughput) network over regular TCP connection.

White box testing of kuksa-databroker using OpenTelemetry in **rust** programming language.

- Accomplished comprehensive observability of the kuksa-databroker by implementing an **OpenTelemetry** pipeline to track function spans, propagate trace contexts, and optimize data collection, which led to a unified view of application performance for analysis in Jaeger.
- Integrated **Prometheus** for metrics collection to monitor service-level performance and expose custom application metrics alongside tracing data.
- Traced **network-level latency** and round-trip times by attaching eBPF probes to kernel socket events and analyzing message delays between microservices.

This observability setup helps in debugging, monitoring performance, identifying latency bottlenecks, and diagnosing failures by giving a comprehensive view of the application's behavior and interactions.

Multi-Platform Framework for Vehicular Functions

Developed a software framework to deploy a single codebase for vehicular functions across simulation, model, and full-scale ROS-based vehicle (FlexCAR), significantly reducing development time.

Developed a VSS-to-ROS communication bridge using a Raspberry Pi to enable non-invasive teleoperation and GStreamer video feedback from a remote-control center.

UNIVERSITY OF STUTTGART — MAY 2023 – JAN 2024

HIWI | *Python, ROS2, MATLAB*

Fault injection and detection in Franka Emika panda arm by making time series analysis of input signals sent to the arm. And bring the arm to halt when a faulty input is detected. Signal processing was done in **MATLAB Simulink** alongside **ROS** package in MATLAB Simulink for communication with robotic arm.

SofDCar Hackathon – Technical Contributor & Organizer: Contributed to the organization of the SofDCar Hackathon Challenge by coordinating technical resources and preparing onboarding materials. Developed and maintained startup scripts and boilerplate code to streamline project initialization.

ENGINECAL TECHNOLOGY PRIVATE LIMITED, BENGALURU — FEB 2022 – FEB 2023

SOFTWARE ENGINEER | *JavaScript, Python,MEAN stack, AWS, Microsoft Azure, Docker, Kubernetes*

Worked as a Product development Engineer mainly focused on DevOps and Full-stack development.

Developed and maintained a microservices architecture for vehicle data processing with **CI/CD**, deployed on **Azure Kubernetes Service (AKS)** and **AWS Elastic Kubernetes Service (EKS)**.

ELLIPSONIC INDIA SOLUTIONS PVT LTD, BENGALURU — APR 2021 - JAN 2022

SOFTWARE ENGINEER | *JavaScript, Python, AWS, MERN stack*

Developed and maintained full-stack web applications using the MERN stack (MongoDB, Express.js, React, Node.js).

Managed cloud infrastructure with **AWS EC2**, **AWS Lambda**, **AWS CloudFront**.

Designed a web application to control AWS instances via **AWS CLI** with real-time status updates of EC2 instances. Used **AWS CloudWatch** and **AWS Lambda** for event-driven automation on AWS infrastructure.

Contributed to building a virtual platform for the **Ministry of Health, Singapore**. The platform included the functionality of sending email which made use of **Amazon Simple Email Service (SES), Amazon CloudWatch, AWS Lambda.**

Performed **data engineering tasks** in Microsoft Azure, working with **Azure Synapse** Analytics to implement **ETL pipelines** using:

- **Azure Data Factory** for orchestrating data workflows,
- **Azure Data Lake Storage** for scalable storage of structured and unstructured data,
- **Synapse SQL & Spark pools** for data transformation and analysis of large datasets in preparation for analytics workflows.

EDUCATION

UNIVERSITY OF STUTTGART — GPA 1.7

M.SC ELECTRICAL ENGINEERING – SMART SYSTEMS

Relevant Coursework: Service Management and Cloud Computing, and Evaluation, Advanced Mathematics for Signal and Information Processing, Deep Learning, Detection and Pattern Recognition, Hardware Platforms and Programming of Embedded Systems, Modeling and Analysis of Automation Systems, Industrial Automation Systems.

BANGALORE INSTITUTE OF TECHNLOGY — GPA 1.5

B.E ELECTRONICS ANS COMMUNICATIONS

SKILLS

Languages: Python, JavaScript, Go, Rust

Technologies & Tools: Docker, Kubernetes, eBPF, AWS, Azure Microsoft, WebRTC, OpenTelemetry, Prometheus, ROS, gRPC, SQL, NoSQL, MERN Stack, MEAN Stack.

LANGUAGE PROFICIENCY

English: C2, **German:** A2 (Learning)

PERSONAL PROJECTS 🔗

VSS-ROS-Bridge , path-finder (wasm), fort-k8s, terraform-ansible-highavailability-cluster